

01 — Excel Baseline (battery 1260 kW, Transit)

Proves · The whole pipeline on the reference plant — every Excel "Load Demands" cell reproduced 1:1.

Mechanics this scenario turns on

- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\sum J + \sum L = \sum H$ — the sea sets the swing, the battery moves the split.
- Peak-shaving H = average × VariationFactor (propulsion 5 %, hotel 2 %). CoverageFactor is a modelling assumption about how much of a swing a battery can realistically catch — propulsion 0.35, hotel 0.05. Both live in `appsettings.json`, not in code.
- Only the **uncovered** part rejoins the demand: `propulsion' = propulsion + L`, `hotel' = hotel + L`. Covered power (J) is never subtracted from anything.
- The shaft generator is filled before any auxiliary starts (the main engine is already turning), and its output is a load **on** that main engine — which is why the ME figure exceeds propulsion. SG capacity scales with the number of running MEs.
- SFOC (g/kWh) depends on **load**: a large 2-stroke burns ~167 at 63 % and over 230 at 5 %. That curve, not the arithmetic, is why running spare engines at low load is expensive.
- Baseline rule: **no battery** → **the worst combination** (`count - 1`); **battery active** → **the third from worst** (`Math.Max(0, count - 3)`). It models what the ship is assumed to do today.
- Battery Benefit runs the pipeline **twice**: world A (demand = average + uncovered L) and world B (demand = average + the full swing H). `Benefit = (FOC_B - FOC_A) × hours`. Both are **optima**; a pinned baseline is ignored. Unticking "Enable Battery" in the UI gives a *third* world — raw demand, swing carried by nobody — which burns less than A and is not the comparison.

Panels described below · Battery Contribution · Power Demands · Baseline · Integration Levels · Battery Benefit

This is the reference card. The other Excel-plant scenarios (02–10, 12–18) describe only what they change and refer back to this one for the rest.

Trust · verified against the reference workbook. These figures are proof.

The workbook's saved scenario: propulsion 11 463 · hotel 3 800 · SM 0 · ME 2×24 000 · SG 3 250 · AE 4×4 000 · Transit 5 000 h · battery 1260/2000 (Transit).

Battery Contribution

Budget 1260. DpReserve/DpDemand/Mission want 0 (no DP, no crane). Then:

Row	H (wants)	I (takes)	J = I×D (covered)	L = H-J (gensets)	Budget after
Propulsion (D=0.35)	11 463×5% = 573.15	573.15	573.15×0.35 = 200.60	372.55	686.85
Hotel (D=0.05)	3 800×2% = 76	76	3.8	72.2	610.85 unused

Tiles: **PS = 204.4** · **SR = 444.7** (573.15+76 = 649.15 total swing = 204.4+444.7 ✓). Budget is beyond saturation — 610.85 kW buy nothing extra.

Power Demands

- Propulsion' = 11 463 + 372.55 = **11 836** · Hotel' = 3 800 + 72.2 = **3 872**
- SG = min(3 872.2, 3 250) = **3 250** · AE = **622** · ME = 11 835.5 + 3 250 = **15 086** (header)
- Energy: ME 15 085.5×5 000 = **75 427 738** · AE 622.2×5 000 = **3 111 000** kWh

Baseline (No iEMS)

Loads per combo (ME load identical for 1-ME rows: 15 085.5/24 000 = **62.9 %**):

ME/SG/AE	AE load	FOC t/h
1/on/1	622.2/4 000 = 15.6 %	2.6572 ← optimal
1/on/2	7.8 %	2.6603
1/on/3	5.2 %	2.6615 ← baseline (3rd-from-worst, battery rule)
1/on/4	3.9 %	2.6621
2/on/0	— (SG 6 500 covers hotel)	2.6975

FOC example: SFOC_ME(62.9 %) ≈ 166.86 → ME 15 085.5×166.86/1e6 = 2.51712 t/h; AE(5.2 %) ≈ 232.0 → 0.14436; sum 2.6615.

- Baseline FOC = 2.6615×5 000 = **13 307.4 t/yr** (ME 12 585.6 + AE 721.8)
- CO2 = 13 307.4 × 3.93267 = **52 333.6** (ME 49 494.9 + AE 2 838.7 — cards must sum to the total)
- Cost = 13 307.4 × 780 = **\$10 379 775**

Integration Levels

- IL1 = 2.6572×5 000 = **13 286.2 t/yr** (AE 700.7) · CO2 **52 250.4** · cost **\$10 363 271**
- **Savings = 13 307.4 – 13 286.2 = 21.2 t/yr (0.2 %) = \$16 504**
- IL2 adds **0**: L2 redistributes hotel between SG and AE — the split (SG maxed at 100 %) is already optimal. IL3 adds **0** here: the ±(500–3.8)×0.8 spike cycle produces no FOC delta at these load points. All three tier chips show 21.2 — correct, not a bug.

Battery Benefit (green badge)

World B (no battery) demand = 11 463+573.15 / 3 800+76 → optimal 2.69198 t/h. Benefit = (2.69198 – 2.65725) × 5 000 = **173.66 t/yr** × 780 = **\$135 452**.

Takeaway: the whole pipeline on the reference scenario — every Excel Load-Demands cell (I/J/K/L rows 8–9) matches the allocation table 1:1.